

Introduction

ar017 · 11 August 2026 · Draft

This collection asks whether a pyramidal–interneuron network gamma (PING) loop can act as a structural sparsity constraint in a task-trained spiking network. The experiments separate three questions that are easy to muddle: what generates the rhythm, what sets the firing-rate floor, and whether the resulting sparse activity remains useful for classification. They also distinguish a genuine experimental dependency—one entry consuming another entry’s checkpoints or measurements—from a looser conceptual dependency.

exp022 — Training-run guide

Link to exp022. The collection’s operational hub owns the canonical training registry and produces the shared COBA and PING checkpoint banks.

Child experiments

1. exp024 — Accuracy converges, firing rate does not

Link to exp024. Audits exp022’s per-epoch baseline histories, comparing convergence of classification accuracy and firing rate. **Manuscript:** supports the rate-attractor interpretation of Figure 3; it is not the plotted image source.

2. exp025 — PING locks E rate $\approx 10\times$ below COBA

Link to exp025. Uses exp022’s trained cells for the central comparison between loop-free COBA and gamma-gated PING. **Manuscript:** Figure 3.

3. exp037 — PING tolerates 80% dropped spikes but collapses on added noise

Link to exp037. Applies spike-deletion and spike-addition perturbations to exp022’s trained cells. **Manuscript:** Figure 8.

4. exp038 — Switching the loop on at inference cuts E rate $\approx 15\times$

Link to exp038. Transfers a trained loop-free baseline into a fresh inhibitory architecture, separating the loop’s inference-time effect from training history. **Manuscript:** Figure 4.

5. exp041 — E rate is affine in gamma frequency

Link to exp041. Uses exp022’s τ_{GABA} family to measure how inhibitory decay changes gamma frequency and excitatory rate. **Manuscript:** supplies the spiking comparison in Figure 2 and the complete Figure 6.

1. exp033 — Mean-field PING

Link to exp033. Compares the measured sweep with the mean-field model.

Manuscript: mean-field panels in Figure 2.

2. exp042 — Timing perturbations

Link to exp042. Uses exp022 baseline checkpoints and exp041 timing measurements to separate inhibitory timing from inhibitory level. **Manuscript:** Figure 9.

3. **exp046 — Per-cycle E-spike count**

Link to exp046. Counts excitatory spikes within cycles measured by exp041.

Manuscript: Figure 7.

6. **exp044 — Rate floor stable across a $20\times \Delta t$ sweep**

Link to exp044. Reads exp022's timestep-sweep cells to test whether the rate floor is physical rather than a step-count artefact. **Manuscript:** Figure 10.

7. **exp048 — Temporal and spatial evidence limits of trained PING**

Link to exp048. Loads the canonical trained PING checkpoint and varies presentation time, input rate, and evidence distribution. **Manuscript:** current source for Figures 11 and 12, pending replacement by exp082.

8. **exp049 — Gradient descent does not preserve a trainable PING loop**

Link to exp049. Uses exp022's initialisation-family cells to test whether training retains or dismantles the recurrent loop. **Manuscript:** Figure 5.

9. **exp082 — Variable-rate streaming with a spike-rate readout**

Link to exp082. Consumes exp022's variable-rate checkpoint bank. Exp080 and exp081 support its design conceptually but are not execution dependencies. **Manuscript:** planned replacement source for Figures 11 and 12 after results are complete.

exp023 — PING fundamentals

Link to exp023. Establishes the free-running excitatory–inhibitory gamma mechanism without training. **Manuscript:** Figure 1.

exp047 — Pool-size invariance requires inverse synaptic scaling

Link to exp047. Separates nominal total coupling from realised per-synapse strength while varying inhibitory-pool size.

exp054 — A PING rhythmicity metric

Link to exp054. Calibrates a rhythmicity statistic and identifies its low-rate and shared-input failure modes using independent simulations. **Manuscript:** coupling-plane and spiking panels in the Figure 2 composite.

exp080 — Empirical input-rate calibration for variable-rate PING training

Link to exp080. Calibrates transformations of sparse pixel intensities into variable Poisson input rates. It conceptually informs exp082 but supplies no runtime artifact.

exp081 — Linear-filter analysis of sparse conductance-driven pixel features

Link to exp081. Analyses how sparse conductance-driven pixel streams are filtered before and within the PING network. It conceptually informs exp082 but supplies no runtime artifact.